Probabilistic Reasoning Homework

Homework (Probability) 1:

Assume a university has undergraduate students and graduate students. The probability distributions are

p_1 = 0.24 for 1st year students,
p_2 = 0.22 for 2nd year students,
p_3 = 0.20 for 3rd year students,
p_4 = 0.18 for 4th year students, and
p_5 = 0.16 for graduate students.

When you see 4 students in the corner, what is
(i) The probability that all of them come from 4th year ?
(ii) The probability that two are from 1st year and two are from 2nd year?
(iii) The combination of student seniorities which achieves the highest probability?
(iv) The combination of student seniorities which achieves the lowest probability?

Note: Need to use Laplace smoothing.

Homework on Probability 2

HW2

A problem on a multiple-choice quiz is answered correctly with probability 0.9 if a student is prepared. An unprepared student guesses between 4 possible answers, so the probability of choosing the right answer is 1/4. Seventy-five percent of students prepare for the quiz. If Mr. X gives a correct answer to this problem, what is the chance that he did not prepare for the quiz?

Homework: Prob3

Mammals Data using Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Assume x has the following attributes. Is it a mannal?

x =

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	yes	yes	yes	?

Homework Probability 4.







Day: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Weather: S S R C C R R S S S R C C R S S C

Given above sequence of 17 weather events, estimate the transition probability A:

$$A_{x_{t-1},x_t} = P(x_t \mid x_{t-1})$$

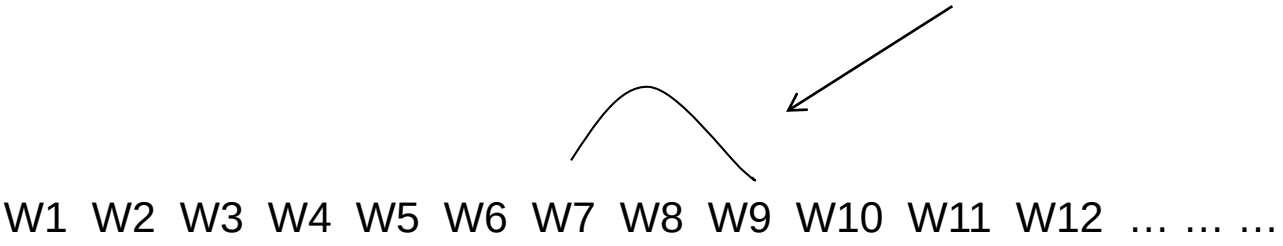
Hint: Draw contingency table. Compute conditional probabilities.

Homework probability 5

What is the probability of 3 contiuous sunny day ?
Assume weather={**S**unny, C**l**oudy , **R**aining} with transition probability matrix:

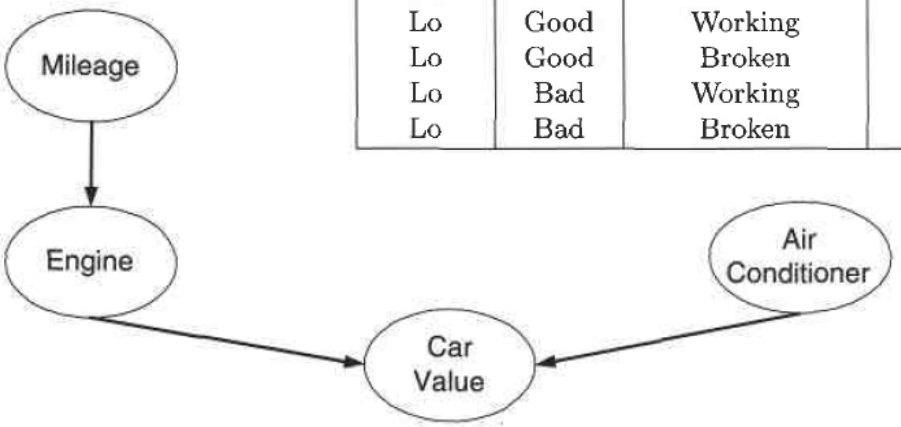
$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

3-day sequence can start any day



Homework: probability 6

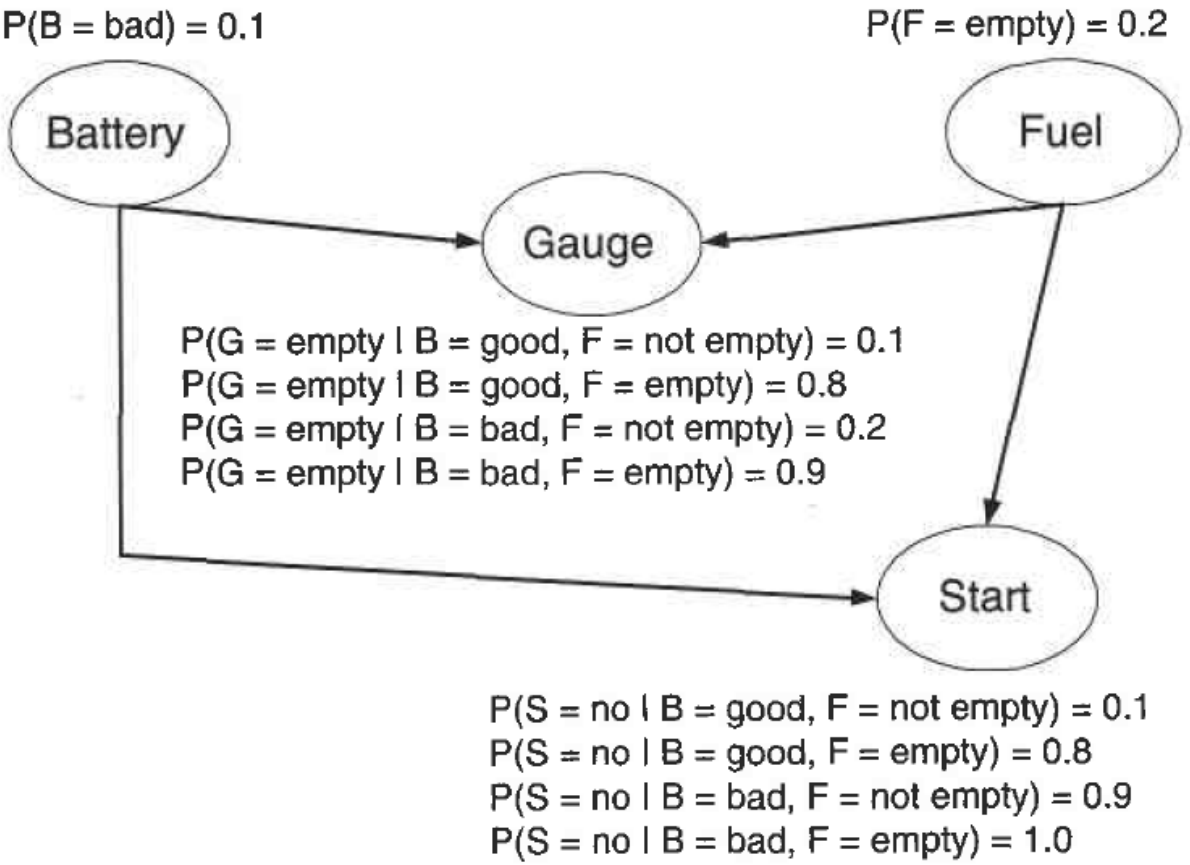
Mileage	Engine	Air Conditioner	Number of Records with Car Value=Hi	Number of Records with Car Value=Lo
Hi	Good	Working	3	4
Hi	Good	Broken	1	2
Hi	Bad	Working	1	5
Hi	Bad	Broken	0	4
Lo	Good	Working	9	0
Lo	Good	Broken	5	1
Lo	Bad	Working	1	2
Lo	Bad	Broken	0	2



This line means there are 3+4 data instances:
3 instances for car-value= High,
4 instances for car-value = Low

Given Bayesian Network, and the training data instances.
Tasks:
1. Write down the probability distribution decomposition.
2. Learn network parameters: P(M), P(A), P(E|M), P(C|E,A).
   Hint: When estimate P(C|E,A), you need to integrate out *M.* Thus there are 3+4+9+0
   - data instances with E=good AND A=working , 3+9 instances for C=Hi, and
   - 4+0 instances for C=low.
3. Compute Prob(Engine = bad, Air Condition = Broken)

Homework Probabilty 7



Tasks:

1. Write down the probability distribution decomposition.
3. Compute:

(a) $P(\mathrm{B} = \text{good}, \mathrm{F} = \text{empty}, \mathrm{G} = \text{empty}, \mathrm{S} = \text{yes})$.

(b) $P(\mathrm{B} = \text{bad}, \mathrm{F} = \text{empty}, \mathrm{G} = \text{not empty}, \mathrm{S} = \text{no})$.

(c) Given that the battery is bad, compute the probability that the car will start.